United International **University**

Department of Computer Science and Engineering

Final Exam, Summer 2025

EEE 4261: GREEN COMPUTING

Total Marks: 40, Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Answer all the questions.

1. ***TechNova Inc.*** is a mid-sized consumer electronics company recently released a line of wireless earbuds and added a small label on the packaging that says: *"Eco-smart packaging: 20% less plastic!"* They promoted this feature in ads and social media but didn't change the product's manufacturing process, energy use, or lifecycle management. The initiative was driven more by the marketing team than by any formal sustainability plan. **5**

a. Based on its approach to sustainability, how would you classify ***TechNova's** business strategy?*

b. What kind of green misrepresentation or false advertizing is ***TechNova*** engaging in, according to the scenario?

2. A small business plans to transition many of its services to the cloud and digitize previously paper-based non-IT functions to save money and improve sustainability. You're asked to create a roadmap. What steps would you recommend to ensure the transition is both cost-effective and green? **5**

Focus on the functions you would digitize first. How would you evaluate when moving to cloud services is advantageous from a cost and energy perspective? **Justify** your answer.

3. ***ABC*** is a startup deciding between purchasing high-end laptops from Supplier A (non-green certified but well-established) or mid-tier laptops from Supplier B that claim lower power-consumption and support take-back/disposal programmes. They also expect rapid turnover of devices in 3 years. As the IT consultant of ***ABC***, how would you advise them on the green computing decision given their business model of frequent refresh and replacement? **5**

4. The IT sector is rapidly expanding: data centers, cloud computing, crypto currency mining, abundance of smart devices, etc. A 2025 report estimates that the global ICT sector could account for **up to 8% of global carbon emissions** by 2030 if no corrective action is taken. You are part of a sustainability task force at a global tech company aiming to align operations that support the 2°C limit. **5**

What concrete steps should your company take to reduce its own impact, and how can it help other industries achieve climate targets through digital innovation?

5. A tech company operates a regional data-center and some remote branch offices. They discover that the vast majority of their IT energy use comes from equipment running 24/7 (even when idle) and the cooling systems in the data-center are outdated. A recent internal **5**

survey shows that the IT division accounts for 30 % of the company's total emissions (electricity use + upstream hardware manufacturing). The VP of IT wants you to propose a pilot project that targets both **Energy Consumption** and **Greenhouse Gas Emissions** reduction. Explain your proposal in detail.

6. After the pandemic, a company shifts to remote work. This reduces commuting but increases household energy use and reliance on cloud storage. **Analyze** how remote working practices align with green computing principles and contribute to the following SDG: **5**

a. Sustainable cities
b. Climate action

7. A growing smart city is facing challenges with excessive energy consumption, water wastage, and poor waste management. The city council decides to implement an IoT-based Green Management System to make the city more sustainable. The system should use sensors and data analytics to monitor and optimize resource usage in real time. As a Computer Engineer, you are tasked with **designing an IoT-based green system** to support the city's sustainability goals. In this connection answer the following questions. **5**

a. What IoT components (sensors, networks, platforms) would you include?
b. How would data be collected, processed, and used for decision-making?

8. A large university plans to build a new data center to support its cloud computing, AI research, and digital learning platforms. However, the administration is concerned about the high energy consumption and carbon emissions associated with traditional data centers. They want the new facility to be environmentally sustainable, energy-efficient, and aligned with the principles of green computing. As the lead system designer, you are asked to propose a **design** plan for a Green Data Center that supports the university's sustainability vision. Based on this scenario, answer the following questions. **5**

a. What key technologies and architectural choices would you include to make the data center energy-efficient?
b. How would renewable energy sources and smart cooling systems be integrated?